Session Three Pop Quiz

- Can you list the Six Steps in performing accounting analysis?

Six Steps in Performing Accounting Analysis

Step 1: Identify Principal Accounting Policies

Step 2: Assess Accounting Flexibility

Step 3: Evaluate Accounting Strategy

Step 4: Evaluate the Quality of Disclosure

Step 5: Identify Potential Red Flags

Step 6: Undo Accounting Distortions

Accounting Adjustments and Recasting

Key Concepts in This Class

- We focus on **Step 6**: accounting adjustments and recasting financial statements
  - Different firms use different accounting methods and estimates
  - Even if two firms with similar economic activities, with different methods and estimates, they may have drastically different financial statements
  - Analysts need to make their own adjustments to make firms' financial statements comparable

Accounting Adjustment Cases

1. Adjusting aggressive revenue recognition
2. Adjusting understated allowances
3. Adjusting delayed asset impairments
4. Adjusting aggressive/conservative depreciation policy
5. Capitalizing R&D expenses
6. Adjusting understated post-employment benefit obligations (qualitative)

1. Adjusting aggressive revenue recognition

- Incentives for aggressive revenue recognition
  - Mainly to increase earnings
- How to detect aggressive revenue recognition
  - Abnormal revenue recognition policy relative to industry peers (Notes)
  - Abnormally high accounts receivable relative to revenue
  - Increases in the gap between earnings relative to operating cash flow (Accruals=Earnings –Operating CF)

Mini Case: MicroStrategy

- **Core Business: Business Intelligence Software**
- **Revenue Streams:**
  - Software Licensing – Subscription-based fees.
  - Support & Maintenance – Annual fees for updates and support.
  - Professional Services – Consulting, custom development, and training.
- **What should be the correct revenue recognition policy?**

Mini Case: MicroStrategy (cont'd)

MicroStrategy

Revenue Recognition:

- Software Licensing subscription-based fees
  - Recognize revenue **over time**, as the customer has access to the software during the subscription period.
- Support & Maintenance – Annual fees for updates and support.
  - Recognize revenue **over the term of the contract**, as the service is provided continuously
- Professional Services – Consulting, custom development, and training.
  - Recognize revenue as services are performed

Mini Case: MicroStrategy (cont'd)

- What was wrong with MicroStrategy's accounting?
  - Recognized revenue upfront rather than over the length of the contract
  - According to the SEC restatement order:

MicroStrategy's software license sales were part of multiple-element transactions that included other services such as product support and development and consulting.

  - MicroStrategy had:
    1. characterized revenue in multiple element transactions as product revenue at the time of the initial transaction
    2. Failed to execute contracts in same period for which revenue was recorded

MicroStrategy

MicroStrategy Inc.

- Restated earnings for the past three years on March 20, 2000

Table 15: Selected Financial Data, 1997-1999

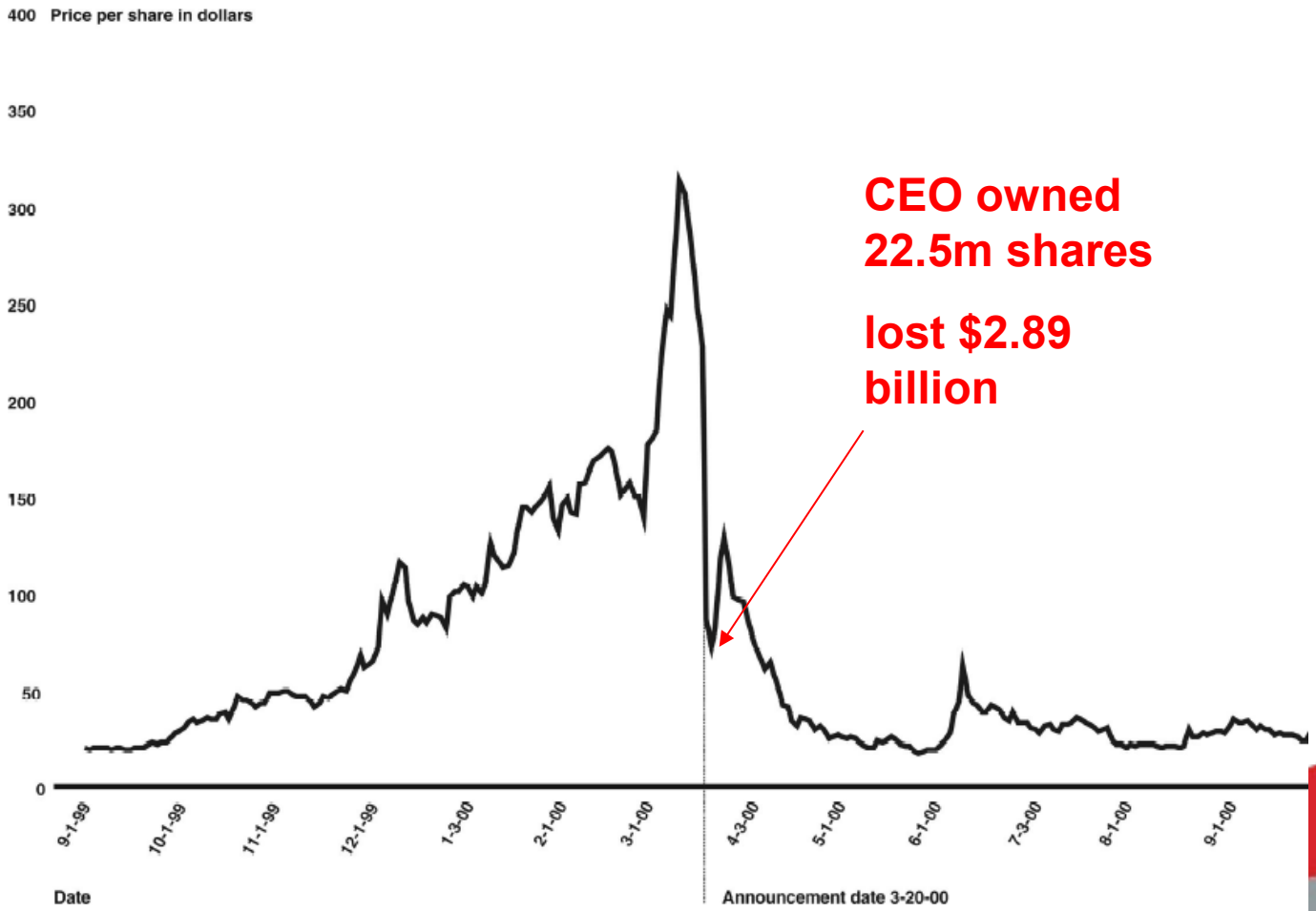
(Dollars in thousands)

Affected financial data	Fiscal year 1997	Fiscal year 1998	Fiscal year 1999
Net income, as reported	$121	$6,178	$12,620
Net income (loss), as restated	(885)	(2,255)	(33,743)

Source: SEC filings.

MicroStrategy Inc.

Figure 18: Daily Stock Prices for MicroStrategy, September 1, 1999–September 29, 2000



Note: Stock prices have been adjusted to account for the January 26, 2000, 2:1 stock split.

Source: GAO's analysis of NYSE TAQ data.

1. Adjusting aggressive revenue recognition - Example (cont'd)

MicroStrategy

MicroStrategy Inc. incorrectly overstated its 1999 revenues by $54.5 million. These revenues were recognized upfront, but instead they should have been recognized over the length of the contract in future periods. Assume all sales were on cash and cost of sales is 3%. Undo the distortion to **1999** financial statements. Ignore tax effect.

1. Adjusting aggressive revenue recognition - Example (cont'd)

MicroStrategy

Wrong accounting:

- Income statement:

1)Revenue ↑ $54.5

2)Cost of sales ↑$1.64 [= $54.5*3%]

3)P&L ↑$52.9

- Balance sheet:

1)Cash↑ $54.5

2)Inventory ↓$1.64

3)S/H equity ↑ $52.9

Correct accounting:

- Income statement:

No impact.

- Balance sheet:

1)Cash↑ $54.5

2)Deferred revenue ↑ $54.5

1. Adjusting aggressive revenue recognition - Example (cont'd)

MicroStrategy

- **Adjustments** to 1999 income statement:

1)Revenue ↓ $54.5

2)Cost of sales ↓$1.64

3)P&L ↓$52.9

- **Adjustments** to 1999 balance sheet:

1)Current liabilities (deferred revenue) ↑ $54.5

2)Current assets (inventory) ↑$1.64

3)S/H equity (retained earnings) ↓ $52.9

Deferred Taxes

- A firm's accounting book and tax book are usually different, i.e., taxes not calculated based on accounting earnings.
  - Pre-tax income different from taxable income

Deferred Taxes

- Accounting rules and tax rules are different
  - Pre-tax profit ≠ Taxable profit

Accounting Book	Tax book
Pre-tax Profit (Profit before taxes, following IFRS)	**Taxable Profit** (amount used to compute income tax payable, following tax code)
Tax expense =Pre-tax Profit * Tax rate (reported on the income statement as an expense)	**Current tax expense** (taxes payable on current year's taxable profit)

Temporary timing differences

- Pre-tax profit ≠ Taxable profit, because we record **accounting revenues and expenses** in a period different from the one used for tax purposes
- Because current temporary differences will result in future differences between accounting and tax profits, they generate deferred tax assets and liabilities

- Deferred tax assets (DTA): you've the right to pay lower income taxes in the future, when the temporary differences reverse.
- Deferred tax liabilities (DTL): you've the obligation to pay extra income taxes in the future, when the temporary differences reverse.

Note: Permanent differences between accounting and tax books do NOT generate DTA or DTL.

Example 1: Unearned revenue

- Receiving cash from customers in advance

Accounting Book		Tax book
Unearned revenue		**Revenue**
Pre-tax Profit	**<**	**Taxable Profit**
Tax expense	**<**	**Current tax expense**

Paying higher taxes now because higher revenue in tax book
⇒ Lower taxes will be paid in future when recording accounting revenue
⇒ Creating DTA

<u>Adjustments:</u> Unearned Revenue↑ => DTA↑ (or DTL↓)

Example 2: Allowance

- Recording allowance before write-off

Accounting Book		Tax book
Bad debt expense		**No expense until direct write-off**
Pre-tax Profit	**<**	**Taxable Profit**
Tax expense	**<**	**Current tax expense**

Paying higher taxes now because lower expenses on tax book
⇒ Lower taxes will be paid in the future when writing off receivables in tax book
⇒ Creating DTA

<u>Adjustments:</u> Allowance↑ (or Net Receivables↓) => DTA↑ (or DTL↓)

Example 3: Depreciation of PP&E

Accounting Book		Tax book
Lower depreciation expense, longer useful life		Higher depreciation expense, shorter useful life
Pre-tax Profit	>	Taxable Profit
Tax expense	>	Current tax expense
Net Book value	>	Net Book value

Paying lower taxes now because higher expenses on tax book
⇒ Higher taxes will be paid in future when PP&E is fully depreciated in tax book
⇒ Creating DTL

<u>Adjustments:</u> NBV↑ (or Accumulated depreciation↓) => DTL↑(or DTA↓)

Notes to Deferred Taxes

- Tax codes differ across jurisdictions
  - Goodwill amortization only taxable in some countries
  - In some countries, tax and accounting books have the same depreciation rate
- A multinational firm may be subject to several different tax codes at the same time
- Our adjustments on DTL/DTA will only be a <u>rough approximation</u>
- For this course, we use the following approximation:
  - **Assets↑ =>DTL ↑ (or DTA ↓)**
  - **Liabilities ↑ =>DTA↑ (or DTL↓)**

1. Adjusting aggressive revenue recognition - Example (cont'd)

MicroStrategy Inc. incorrectly overstated its 1999 revenues by $54.5 million. These revenues were recognized upfront, but instead they should have been recognized over the length of the contract. Assume all sales were on cash and cost of sales is 3%. Undo the distortion to 1999 financial statements. Assuming 35% tax rate and the firm as deferred tax liability on its balance sheet.

1. Adjusting aggressive revenue recognition example (cont'd)

MicroStrategy

- Adjustments to 1999 income statement:

1)Revenue ↓ $54.5

2)Cost of sales ↓$1.6 [= 54.5*3%]

Pre-tax profit ↓$52.9 [=(54.5-1.6)]

3)Tax expense ↓ $18.5 [=52.9*35%]

4)P&L ↓$34.4

- Adjustments to 1999 balance sheet:

1)Current liabilities (unearned revenue) ↑ $54.5

2)Current assets (inventory) ↑$1.6

3) DTL↓$18.5 (or DTA ↑ $18.5) [= (54.5-1.6)*35%]

4)S/H equity ↓ $34.4

1. Adjusting aggressive revenue recognition - Exercise

In early 2023 Bristol-Myers Squibb announced that it would have to restate its financial statements as a result of stuffing as much as $3.35 billion worth of products into wholesalers' warehouses from 2019 through 2021. The company's sales and cost of sales during this period was as follows:

Bristol Myers Squibb

(in $ millions)	**2021**	**2020**	**2019**
Net sales	18,139	17,695	16,502
Cost of goods sold (COGS)	5,454	4,729	4,458

The company's marginal tax rate during the three years was 35%. What adjustments are required to correct Bristol-Myers Squibb's income statement and balance sheet for December 31, 2021? What assumptions underlie your adjustments?

1. Adjusting aggressive revenue recognition - Exercise

Bristol Myers Squibb

Assumption 1: All adjustments will be reflected in 2021 financial statements.

Assumption 2: The adjusted inventories/COGS reflect the weighted average COGS as a percentage of revenue during 2019-2021.

COGS/Revenue=(5,454+4,729+4,458)/(18,139+17,695+16,502)=28%

Adjustments to income statement of 2021 (in $billions):

- Net sales ↓ $3.35
- Cost of goods sold ↓ $0.94 (=3.35*28%)
- Pre-tax profit ↓ $2.41
- Tax expense ↓ $0.84 (=2.41*35%)
- P&L ↓ $ 1.57

Adjustments to balance sheet of 2021 (in $billions):

- Accounts receivables ↓ $3.35
- Inventories ↑ $0.94
- Deferred tax liability ↓ $0.84
- Shareholders' equity ↓ $1.57

1. Adjusting aggressive revenue recognition - Summary

- Impact on financial statements:
  - Overstated P&L and shareholders' equity.
  - Overstated assets (accounts receivable), if falsely creating revenue.

  Or Understated liabilities (deferred/unearned revenue), if falsely recognized revenue too early.

- Adjustments:
  - Reduce overstated revenue, P&L and shareholders' equity
  - Reduce overstated accounts receivable

  Or Increase understated deferred revenue
  - Add back cost of goods sold and inventories
  - Don't forget to adjust tax expense and DTA/DTL

2. Adjusting understated allowances

- Incentives for understating allowances
  - Mainly to increase assets and decrease expenses
- How to detect understated allowances
  - Unexplained/Unreasonable changes in a firm's allowance for doubtful accounts (Notes)
  - The size of allowance lower than industry practice
  - Recognized allowances systematically lower than its write-offs

2. Adjusting understated allowances - Example

What concerns might investors have about allowance accounts?

The Motley Fool
To Educate, Amuse & Enrich™

FOOL ON THE HILL

Accounting for Non-Paying Customers

By Whitney Tilson
January 9, 2001

Gateway

Over the past eight quarters, Gateway's allowance for doubtful accounts has been the following:

Quarter	Allowance
Q4 98	2.61%
Q1 99	2.17%
Q2 99	2.70%
Q3 99	2.95%
Q4 99	2.49%
Q1 00	2.46%
Q2 00	1.99%
Q3 00	2.00%

Why is this lower than historical average?

What inferences might one draw from the pattern shown in the article?

Gateway, Inc.'s allowance for doubtful accounts

A decrease by 0.47% might not sound like much, but…

- Gateway met analysts' estimates of $0.46 per share for the quarter.
- Without this adjustment, it would have reported $0.44 (a miss)

Response from Gateway:

"Our percentage is in line or higher than competitors. Second, we have taken measures to reduce [the] allowance. The two key things we have done is that we have put in a predictive dialer process to automate and ensure we contact delinquents accounts and second, we have increased our aggressiveness against our larger customers which has significantly improved our delinquents. Additionally, in Q3 the timing of business sales favorably impacted [the] reserve %. Last, this does not include reserve on bad debt for consumer loans."

Does it seem reasonable?

Gateway, Inc.'s allowance for doubtful accounts

Quarter	Allowance
Q3 00	2.00%
Q4 00	2.28%
Q1 01	3.07%
Q2 01	3.01%

2. Adjusting understated allowances - Example

- Recall Gateway, Inc.

Gateway

Over the past eight quarters, Gateway's allowance for doubtful accounts has been the following:

Quarter	Allowance
Q4 98	2.61%
Q1 99	2.17%
Q2 99	2.70%
Q3 99	2.95%
Q4 99	2.49%
Q1 00	2.46%
Q2 00	1.99%
Q3 00	2.00%

Let's assume we decide to apply 2.46% for Q2 of 2000.

Gateway

Gateway disclosed the following information on its allowance and accounts receivables in its financial statements :

(in 000' dollars)	Gross receivables	Allowance	Allowance /Gross receivables
Q4 99	662,811	16,472	2.49%
Q1 00	668,502	16,475	2.46%
Q2 00	672,054	13,377	~~1.99%~~ 2.46%

If we apply 2.46% for Q2 of 2000, undo the distortion to 2000-Q2 financial statements. Gateway has a tax rate of 38%.

<u>Adjustments:</u>

Allowance↑3,156 [=(2.46%-1.99%)*672,054]

0.47%

Bad debt expense ↑3,156

2. Adjusting understated allowances - Example (cont'd)

- <u>Adjustments to income statement:</u>

1)Bad debt expense ↑ $3,156 (=>Pre-tax profit ↓$3,156)

2)Tax expense ↓ $1,199 [=$3,156*38%]

3)P&L ↓$1,957

- <u>Adjustments to balance sheet:</u>

1)Net receivables ↓$3,156 (Allowance ↑ $3,156)

2)DTA ↑$1,199 (or DTL↓$1,199)

3)S/H equity ↓ $1,957

2. Adjusting understated allowances - Summary

- Impact on financial statements:
  - Overstated assets (net accounts receivables)
  - Understated expenses (bad debt expense)
- Adjustments:
  - Reduce overstated assets
  - Increase understated expense
  - Don't forget to adjust tax expense and DTA/DTL

3. Adjusting delayed asset impairments

- Incentives for delaying impairments
  - Mainly to increase assets and decrease expenses
- How to detect delayed impairments
  - Whether industry peers have recently had asset impairments, suggesting an economy/industry-wide downturn
  - Whether the firm has a history or regular write-downs, suggesting a tendency to delay

3. Adjusting delayed asset impairments - Example

EasyJet failed to impair its PP&E of £72.4 million in 2020. Assuming its tax rate is 28%. Undo this distortion to its 2020 financial statements.

3. Adjusting delayed asset impairments - Example

easyJet

- Adjustments to income statement:

1) Other operating expense ↑ £72.4 (=>Pre-tax profit ↓ £72.4)

2) Tax expense ↓ £20.3 [= £72.4*28%]

3) P&L ↓£52.1

- Adjustments to balance sheet:

1) Non-current fixed assets↓£72.4

2) DTL↓£20.3 (or DTA ↑ £20.3)

3) S/H equity ↓ £52.1

3. Adjusting delayed asset impairments - Summary

- Impact of delayed asset impairments
  - Overstated assets
  - Understated expenses
- Adjustments:
  - Reduce overstated assets
  - Increase understated expense
  - Don’t forget to adjust tax expense and DTA/DTL

4. Adjusting depreciation policy

- Recall common depreciation methods
  - Straight-line (most common)
  - Declining balance
  - Unit of production

4. Adjusting depreciation policy (cont'd)

- Recall key estimates to calculate depreciation
  - Estimated useful life; Salvage (residual) value
- Different estimates lead to substantial differences in balance sheet and income statement, even for similar economic activities, especially for industries with heavy depreciation expenses (e.g. airlines and utilities)
- How to detect abnormal depreciation policy
  - Whether these two estimates consistent with industry peers
  - Whether the firm changed estimates without justification

4. Adjusting depreciation policy (cont'd)

- Adjustments:
  - Use industry's depreciation rates to recalculate depreciation expense
  - We need to apply changes retroactively, i.e. to recognize <u>cumulative effects over the asset's life to date</u>

4. Adjusting depreciation policy - Example

- In 2011, Lufthansa straight-line depreciated its aircraft over an estimated useful life of 12 years and assumed a salvage value of 15% of its initial cost. Industry peers such as Air France-KLM and British Airways used estimated useful life of 20 years and salvage value of 10%. On Dec. 31, 2011, Lufthansa reported its fleet of aircraft had original cost of €22,484 million, and accumulated depreciation of €12,238. Lufthansa has a tax rate of 30%.
- Recast Lufthansa's 2011 balance sheet and income statement using industry depreciation estimates.

4. Adjusting depreciation policy - Example

Lufthansa

1) Annual depn expense (old)= €22,484 *[(1-15%)/12]= €1,593

2) Annual depn expense (new)= €22,484 *[(1-10%)/20]= €1,012

3)Average asset life=Accumulated Depn (old)/Annual depn expense (old)= €12,238/€1,593=7.682 years

4)Accumulated Depn (new)=Average asset life * Annual depn expense (new)=7.682* €1,012= €7,775

4. Adjusting depreciation policy example (cont'd)

- Adjustments to income statement:

1)Depreciation expense ↓ €581 [=1,593-1,012]

2)Tax expense ↑ €174.3 [= 581*30%]

3)P&L ↑ €406.7

- Adjustments to balance sheet:

1)Accumulated depreciation ↓ € 4,463 [=12,238-7,775]

2)NBV of assets ↑ € 4,463

3)DTL↑ € 1,339 [=4,463*30%]

4)S/H equity (retained earnings)* ↑ €3,124 [=406.7* 7.682years]

* S/H equity change reflects the cumulative changes in P&L in previous years over asset's life to date

4. Adjusting depreciation policy - Summary

- Effects of a lower/aggressive depreciation policy relative to a higher/conservative depreciation policy

1) Earnings is higher in initial years
2) Net book value (NBV) of assets and shareholders' equity are higher
3) Cash flows is unaffected

- Incentives for an aggressive depreciation policy
  - Overstated earnings and NBV of assets
- Incentives for a conservative depreciation policy
  - Higher depreciation expense saves tax in some countries

5. Capitalizing R&D expenses

- Reasons for recognizing R&D as expenses rather than intangible assets
  - Required by accounting standards
- How to detect R&D expenses
  - Disclosed as a line item in income statement usually for research-intensive firms

5. Capitalizing R&D expenses (cont'd)

- Impact of expensing R&D on financial statements:
  - Understated intangible assets
  - Overstated expenses

- Adjustments:
  - Treating R&D expenses as purchases of intangible assets
  - Specify an amortizable life for R&D (e.g. 5 years)
  - Collect information on past R&D expenses within the amortizable life (e.g. 5 years)
  - Add up all the unamortized R&D for the current period, i.e. apply changes retroactively and recognize cumulative effects over the asset's life to date

5. Capitalizing R&D expenses - Example

SAP has the following R&D expenses between 2015-2020. Recast its 2020 financial statements after capitalizing all the R&D expenses. Assume:

1) All R&D expenses incurred at the end of each year.
2) 5 years amortizable life.
3) Tax rate is 36.5%.

in million Euros	
Year	R&D expense
2020	1,020.02
2019	993.99
2018	909.39
2017	898.25
2016	969.38
2015	744.67

Year	R&D expense	Unamortized portion	Unamortized value	Amortization for 2020	
2020	1,020.02	100%	1,020.02		
2019	993.99	80%	795.19	198.80	=20%*993.99
2018	909.39	60%	545.63	181.88	=20%*909.39
2017	898.25	40%	359.30	179.65	=20%*898.25
2016	969.38	20%	193.88	193.88	=20%*969.38
2015	744.67	0%	0.00	148.93	=20%*744.67
Current value of R&D assets:			2,914.02		
Total amortization expense for 2020:				903.14	

- <u>Adjustments to 2020 income statement:</u>

1)R&D expense ↓ €1,020

2)Amortization expense ↑ €903

} **Pre-tax profit ↑ €117 [=1,020-903]**

3)Tax expense ↑ € 43 [= €117 *36.5%]

4)P&L ↑ €74

- <u>Adjustments to 2020 balance sheet:</u>

1)Intangible assets↑ €2,914

2)DTL↑€1,064 [=2,914*36.5%]

3)S/H equity* (retained earnings)↑ €1,850

* S/H equity change reflects the cumulative changes in P&L in previous years over asset's life to date

6. Adjusting understated post-employment benefit obligations

- Post-employment benefit (pension) obligations: present value of future expected payouts
- Post-employment benefit (pension) assets: fair value of funds set aside to meet these obligations
- The net obligations over assets is put at balance sheet (non-current liabilities)
- Pension expenses (P&L) include current service cost, past service cost, and interest cost on net pension obligations
- Actuarial gains and losses and excess return on pension assets go to other comprehensive income (OCI)

6. Adjusting understated post-employment benefit obligations (cont'd)

- Incentive to understate pension obligations
  - Understate liabilities
- How to detect understated post-employment obligations
  - Understand the sensitivity of obligations to various estimates used, e.g. discount rate, assumptions about increase in wage and benefit costs (disclosed in Notes)
- Impact on financial statements
  - An increase in discount rate decreases liabilities
  - Understated future wage benefit rates understate liabilities

Note

- We only make adjustments to income statement and balance sheet, but NOT to cash flow statement, because
  - These adjustments do NOT affect total cash
  - But they may affect **classification of cash** into operating, investing, and financing activities
  - Once we have recast income statement and balance sheet, we will naturally recast cash flow statement

Conclusions

- Recasting financial statements is an important step to facilitate comparability among financial statements analyzed.
- Analysts should focus on evaluating and adjusting accounting measures that describe the firms' key strategic value drivers.
- Should focus on adjustments with a material effect on firm's financial ratios.
- Keep in mind that many accounting adjustments will be estimates.